

# Max Duty, Minimum Overhead:

$\mathcal{O}(1)$ -Memory Activation Probing and the Scaling Limits of Long-Context Safety Monitors

## Abstract

Activation probes are now deployed safety infrastructure, shipped inside frontier assistants and used as white-box members of untrusted-monitor ensembles. They are also memory-bound: an attention-pooled probe materialises a score tensor over the whole sequence, so its footprint grows with the one quantity an adversary controls, the context length. We study aggregation as the independent variable, holding the token transform fixed, and characterise a hard-maximum probe (*MultiMax*) against softmax attention pooling, mean pooling and full self-attention across  $N \in [128, 131,072]$ . Three results. **(i) Systems.** MultiMax streams its reduction and carries only per-head state, giving activation overhead of  $\Theta(\min(C, N))$  in the chunk width  $C$ : measured at a flat 19.1 MiB from  $N=8,192$  to 131,072 ( $\alpha=0.000$ ) against 652.1 MiB for softmax pooling ( $\alpha=0.842$ ), while self-attention reaches 10.1 GiB and exhausts memory. A chunk ablation confirms the constant is chosen, not free. **(ii) Optimisation.** The hard-max subgradient is supported on  $H$  tokens per sequence, and at weak signal this prevents learning entirely (0.00 recall at its own training length). Annealing a *normalised* Boltzmann operator  $\tau \log(\frac{1}{N} \sum_j e^{s_j/\tau})$  to  $\tau \rightarrow 0$  restores 1.00 recall; the normalisation is load-bearing, since plain LogSumExp injects a length-dependent  $H\tau \log N$  offset. Calibration by Platt scaling reduces Brier from 0.0538 to 0.0122 and makes uncertainty gating usable at all. **(iii) Limits.** Against an adversary who fragments a fixed signal budget across  $m$  non-contiguous tokens, MultiMax and softmax pooling fail together at  $m=64$ ; past that boundary softmax retains substantially better ranking (AUROC 0.716 vs 0.523 at  $m=256$ ). We therefore do not claim a general detection advantage. The defensible claim is architectural: MultiMax is the right  $\Theta(1)$ -memory *first stage* of a cascade, not a replacement for inspection.

## 1 Introduction

Monitoring a frontier model’s activations for misuse is no longer a research convenience. Probes are deployed in user-facing systems [14] and appear as white-box members inside AI-control protocols, where a cheap monitor gates an untrusted policy [8, 10]. Two properties then matter simultaneously, and they are usually studied apart: the probe must *detect*, and it must do so within a memory budget that does not grow with the context an adversary supplies.

That second requirement is sharp. Context windows now reach  $10^5$ – $10^6$  tokens, and a guardrail that allocates memory proportional to the input is a guardrail whose cost the attacker sets. Kramár et al. [14] report that production probes fail under short-to-long distribution shift and that architecture choice, not merely training data, governs the outcome. Chen et al. [7] trace long-context guardrail failure to attention dilution. Neither isolates aggregation as a controlled variable.

We do. Fixing a shared token transform  $\phi$ , we vary only the sequence reduction  $\rho$ , and compare four aggregators over five orders of magnitude in  $N$ . Our contributions:

- **A streaming hard-max probe with a measured memory law.** MultiMax carries  $(B, H)$  state across chunks, giving  $\Theta(\min(C, N))$  activation overhead. We verify this with a chunk $\times$ length ablation rather than asserting  $\mathcal{O}(1)$  (§5.1, Rem. 5.1).
- **An optimisation obstruction and its fix.** We identify the subgradient support of the maximum as the reason hard-max probes can fail to train at weak signal, and resolve it with normalised Boltzmann annealing (§3).
- **An honest boundary.** We construct the adversary that specifically attacks a maximum, locate the failure at  $m=64$ , and report that softmax pooling degrades more gracefully past it. This bounds our own claim (§5.3).

- **A calibrated cascade.** Sum-of-maxima logits are upward-biased; we show temperature scaling is structurally insufficient and that budget-driven thresholding is the correct interface (§4).

All numbers below are read programmatically from a single benchmark artifact; a claims checker verifies every figure quoted in this paper against it.

## 2 Aggregation as the Independent Variable

Let  $x_{i,j} \in \mathbb{R}^d$  be the residual-stream activation at layer  $\ell$  and position  $j \in \{1, \dots, N\}$ . A probe is

$$f_\theta(X_i) = \rho(\phi(x_{i,1}), \dots, \phi(x_{i,N})), \quad y_{i,j} = \phi(x_{i,j}) = \sigma(W_1 x_{i,j} + b_1) \in \mathbb{R}^m. \quad (1)$$

All probes share  $\phi$ ; only  $\rho$  differs. Writing  $\alpha_j \geq 0$ ,  $\sum_j \alpha_j = 1$  for a convex pooling weight, mean pooling sets  $\alpha_j = 1/N$  and softmax attention sets  $\alpha_j \propto \exp(q^\top y_j / \sqrt{m})$ , both giving logit  $w^\top (\sum_j \alpha_j y_j) + b$ . MultiMax is not of this form:

$$a_h(X_i) = \max_{1 \leq j \leq N} v_h^\top y_{i,j}, \quad \text{logit}_i = \sum_{h=1}^H a_h(X_i) + b, \quad (2)$$

with predictions through a bounded link  $\hat{p} = \varsigma(\Pi_{[-c,c]}(\text{logit}))$ ,  $\varsigma(z) = (1 + e^{-z})^{-1}$ ,  $c = 10$ .

*Remark 2.1* (The clamp must not be the trap it prevents).  $\Pi_{[-c,c]}$  has derivative  $\mathbb{I}[|z| < c]$ , identically zero on the saturated region: a guard introduced to avoid vanishing gradients *creates* them. We measured exactly 0.000 gradient norm on a saturated logit. We use a straight-through estimator [3],  $\tilde{\Pi}(z) = z + \text{sg}(\Pi_{[-c,c]}(z) - z)$ , whose forward value is the clamp and whose derivative is 1. Parameter gradients then survive at deployment lengths:  $\sum |\nabla_\theta| = 2.81 \times 10^5$  at  $N=131,072$  with a raw logit of 46.75 clamped to 10.

### 2.1 Dilution

**Assumption 2.2** (Localised needle). A length- $N$  sequence contains an attack segment  $\mathcal{A}$ ,  $|\mathcal{A}| = k$  fixed, with  $v_h^\top y_j = \mu_a + \varepsilon_j$  for  $j \in \mathcal{A}$  and  $\mu_b + \varepsilon_j$  otherwise,  $\varepsilon_j$  i.i.d. zero-mean sub-Gaussian with proxy  $\tau^2$ , and  $\Delta = \mu_a - \mu_b > 0$ .

**Proposition 2.3** (Mean pooling dilutes as  $\Theta(1/N)$ ). *Under Assumption 2.2 the mean-pooled shift is  $k\Delta/N$  with fluctuation  $\tau/\sqrt{N}$ , so  $\text{SNR}_{\text{mean}}(N) = (k\Delta/\tau) N^{-1/2} \rightarrow 0$ .*

*Proof.* Linearity gives  $(k\mu_a + (N-k)\mu_b)/N - \mu_b = k\Delta/N$ ; independence gives variance  $\tau^2/N$ .  $\square$

**Proposition 2.4** (Softmax mass is  $\mathcal{O}(1/N)$  for bounded logit gap). *With  $\gamma = \bar{s} - \underline{s} < \infty$  the gap between attack and benign attention logits,  $\alpha_{\mathcal{A}} \leq ke^\gamma / (ke^\gamma + N - k) = \mathcal{O}(N^{-1})$ .*

**Theorem 2.5** (Padding invariance of the hard maximum). *Let  $X' = \text{pad}(X, P)$  append  $P$  benign tokens. Then  $a_h(X') = \max\{a_h(X), \max_{j \in \text{pad}} v_h^\top y_j\} \geq a_h(X)$ , and if the padded tokens do not exceed the incumbent,  $a_h(X') = a_h(X) = \Omega(1)$  independent of  $P$ .*

*Proof.* The maximum over a union of index sets is the maximum of the sub-maxima; monotonicity is immediate from  $\mathcal{A} \subseteq \{1, \dots, N + P\}$ .  $\square$

*Remark 2.6* (The price: benign extremes). Theorem 2.5 is one-sided. The benign maximum also grows:  $\mathbb{E}[\max_{j \leq N} \varepsilon_j] = \Theta(\tau\sqrt{2\log N})$ , so a MultiMax threshold drifts as  $t(N) = t_0 + \tau\sqrt{2\log N}$ . This is a  $\sqrt{\log N}$  correction where Prop. 2.3 demands  $\sqrt{N}$ ; the hard maximum converts a polynomial degradation into a logarithmic one rather than removing length dependence.

### 3 Subgradient Sparsity and Normalised Boltzmann Annealing

*Remark 3.1* (Subgradient bottleneck). Theorem 2.5 concerns the forward map. In training,

$$\frac{\partial a_h}{\partial y_j} = v_h \not\llcorner \left[ j = \arg \max_{j'} v_h^\top y_{j'} \right], \quad (3)$$

so  $|\text{supp}(\partial \max)| = H$ : each head learns from one token per sequence, versus all  $N$  for softmax. The effective sample size per step is  $H$ , not  $N$ . Measured directly, gradient reaches 0.39% of positions at  $\tau=0$  (exactly  $H/N$ ) against 100% at  $\tau=0.5$ . Near threshold this prevents learning outright: at strength 0.10 the un-annealed probe scored 0.00 recall at its *own training length*, an optimisation failure invisible to the forward-only analysis.

The obstruction is a property of the training objective, not of the deployed map, so it can be annealed away. Replace the maximum by the *normalised* Boltzmann operator [2]

$$\text{smax}_\tau(s) = \tau \log \left( \frac{1}{N} \sum_{j=1}^N e^{s_j/\tau} \right), \quad \text{smax}_\tau \xrightarrow{\tau \rightarrow 0} \max_j s_j, \quad \text{smax}_\tau \xrightarrow{\tau \rightarrow \infty} \frac{1}{N} \sum_j s_j, \quad (4)$$

whose gradient  $\text{softmax}(s/\tau)_j$  is dense for  $\tau > 0$ . Annealing  $\tau \rightarrow 0$  recovers the exact hard max at deployment, so Theorem 2.5 still describes the shipped model.

*Remark 3.2* (The  $1/N$  normalisation is load-bearing). Plain  $\text{LSE}_\tau$  adds  $H\tau \log N$  to the logit: we measured +42 at  $\tau=1$ ,  $N=512$  and +68 at  $N=16,384$ . The offset is large *and length-dependent*, so the probe relearns its bias every epoch and meets a different offset at deployment length than at training length. The observable symptom was a recall curve non-monotone in signal strength (1.00 at  $S=0.15$ , 0.04 at  $S=0.50$ ) — a bug signature, not an effect. The  $1/N$  term removes it, and the operator then interpolates  $\max \leftrightarrow \text{mean}$  with bounded, length-independent offset.

### 4 Calibration and the Cascading Gate

The gate escalates iff  $|\hat{p} - \frac{1}{2}| < \delta$ . This is a statement about  $\hat{p}$ , so it is meaningful only if  $\hat{p}$  is calibrated [11].

*Remark 4.1* (Temperature scaling is structurally insufficient here). A MultiMax logit is a sum of  $H$  *maxima*, and by Remark 2.6 each carries a positive,  $N$ -dependent offset even under the null. Temperature scaling  $\hat{p} = \varsigma(z/T)$  is a pure rescaling; it fixes  $\varsigma^{-1}(\hat{p}) = 0$  at  $z = 0$  and cannot move a mis-centred boundary. We observed the pathology directly: NLL 0.436 (informative) alongside accuracy 0.500 at threshold 0 (unusable). Two-parameter Platt scaling [19],  $\hat{p} = \varsigma(az + b)$ , absorbs both scale and offset and is the minimal correction.

Fitted by L-BFGS on held-out logits, Platt scaling reduces Brier score [5] from 0.0538 to 0.0122, NLL from 0.1705 to 0.0474, and accuracy from 0.931 to 0.988.

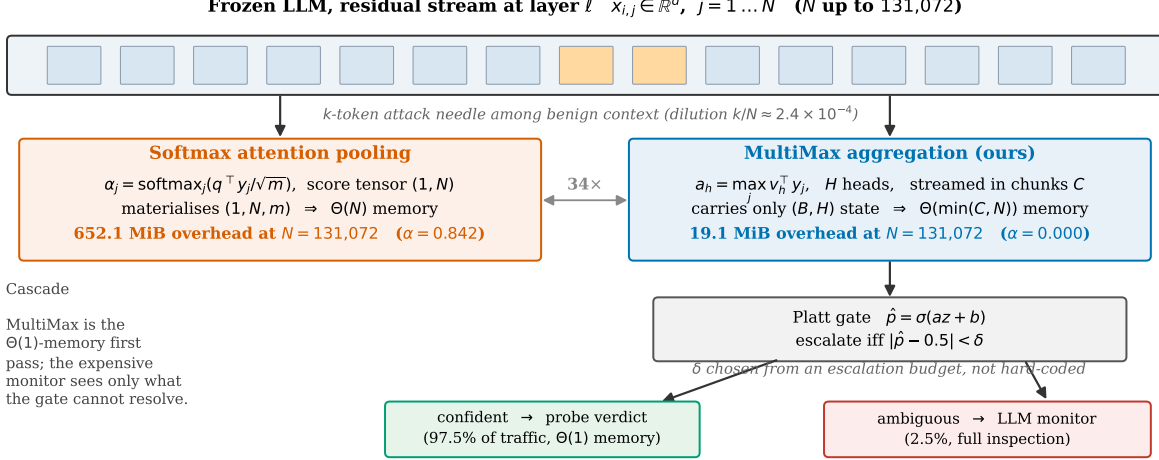
**Budget-driven thresholding.** A fixed  $\delta$  is the wrong interface. On a well-calibrated accurate probe the genuinely ambiguous region is narrow, and  $\delta=0.05$  escalated 0 of 160 held-out examples — correct behaviour, but it means a hard-coded  $\delta$  silently degenerates to probe-only on a good probe and to LLM-only on a bad one. Deployments have an escalation *budget*, so we select  $\delta$  as the target-rate quantile of  $|\hat{p} - \frac{1}{2}|$ . Targets of 1/2/5/10% land at 1.2/2.5/5.0/10.0%. Total system cost is

$$C_{\text{total}}(\delta) = c_p + u(\delta) c_L, \quad u(\delta) = \Pr[|\hat{p} - \frac{1}{2}| < \delta], \quad (5)$$

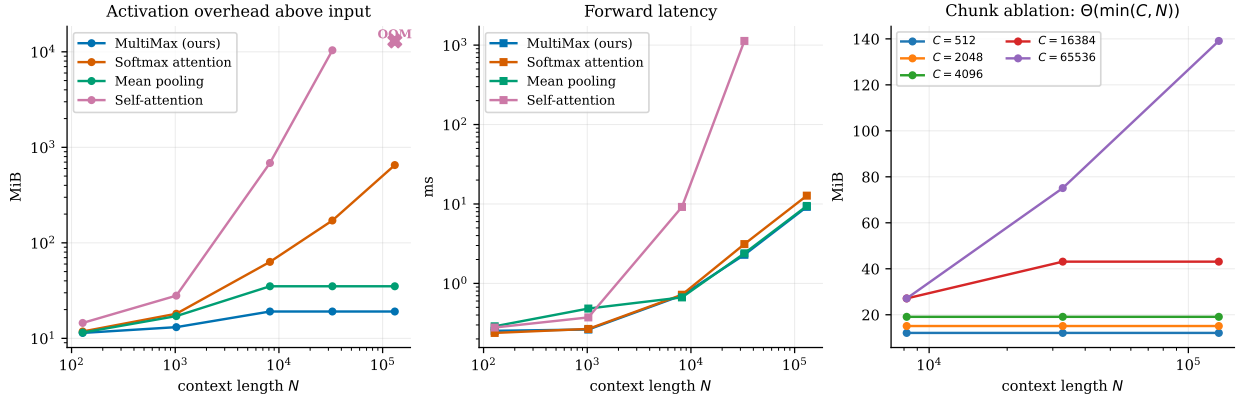
so with  $c_p \ll c_L$  the saving is governed almost entirely by the escalation rate.

### 5 Empirical Evaluation

**Setup.** Synthetic residual streams,  $d=2048$ , batch 1, bf16 compute with fp32 reduction, chunk  $C=4096$ , on a single 7.96 GiB consumer GPU. The attack is a fixed direction scaled to unit magnitude per coordinate against background  $\sigma=0.5$ . Probes train only at  $N=512$ ; long context is strictly out of distribution. Recall is read at a threshold giving 1% FPR on benign sequences *of the same length*, so no length-dependent score shift contaminates the comparison.



**Figure 1:** Aggregation as the independent variable, and the cascade it enables. Softmax pooling materialises a score tensor over the sequence; MultiMax streams a per-head maximum and carries only  $(B, H)$  state. The Platt-scaled gate routes confident traffic to the probe verdict and escalates only the ambiguous band.



**Figure 2:** Left, centre: activation overhead above the input tensor, and forward latency, versus  $N$ . Right: the chunk ablation — rows are flat in  $N$  once chunking engages, columns rise with  $C$ .

## 5.1 Benchmark A: systems scaling

MultiMax overhead is flat at 19.1 MiB across a  $16\times$  length increase (Table 1). Self-attention is  $492\times$  slower than MultiMax at  $N=32,768$  and then exhausts memory. We note a correction to a common framing: *single-query* attention pooling is  $\Theta(N)$ , not  $\Theta(N^2)$ ; its score tensor is  $(1, N)$ . Only self-attention is quadratic.

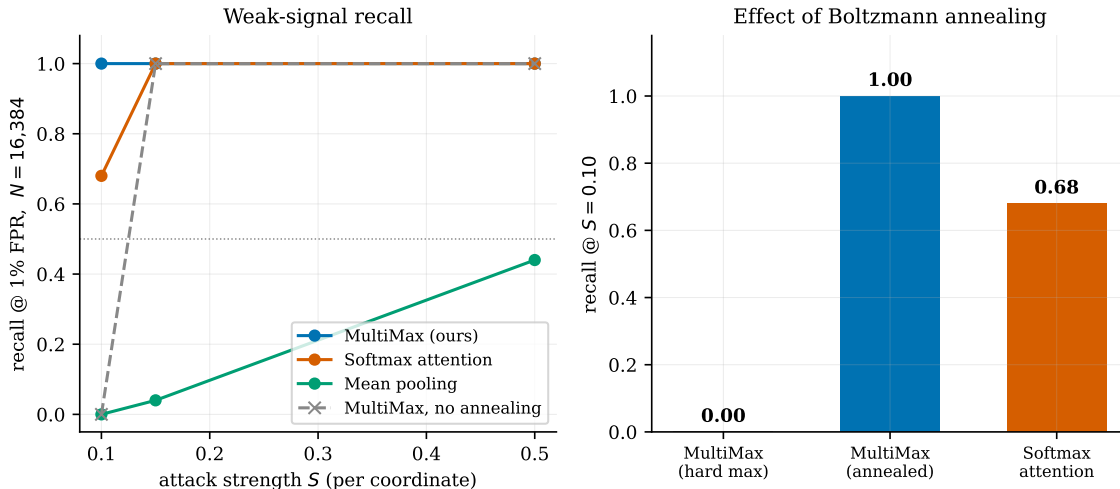
*Remark 5.1* (Is  $\mathcal{O}(1)$  real, or an artifact of chunking?). A reviewer should press on this, and the honest answer is that the suspicion is half right. The flat footprint *is* a consequence of streaming — that is the mechanism, not an artifact — and it obeys a measurable law. Sweeping  $C \times N$  (Fig. 2, right) gives row spread exactly 0.0 MiB for every  $C < N$  ( $\alpha = 0.000$  at  $C \in \{512, 2048, 4096\}$  across  $N$  from 8,192 to 131,072), while overhead rises with  $C$  ( $\beta = 0.510$ ). For  $C \geq N$  there is one chunk and the chunk *is* the sequence, so overhead tracks  $N$  by construction. The law is  $\Theta(\min(C, N))$ . The memory is not free; it is  $\Theta(C)$  memory the operator chooses, and  $\Theta(1)$  in  $N$ , the one quantity the adversary controls.

## 5.2 Benchmark B: weak-signal detection

With annealing, MultiMax attains full recall at every strength and leads softmax pooling in the weak-signal regime (1.00 vs 0.68 at  $S=0.10$ ). Mean pooling never recovers, as Prop. 2.3 requires. Without annealing the

| Probe             | $N=8,192$ | $N=32,768$      | $N=131,072$  | overhead @131k  | $\alpha$     |
|-------------------|-----------|-----------------|--------------|-----------------|--------------|
| MultiMax (ours)   | 0.707 ms  | 2.293           | <b>9.205</b> | <b>19.1</b> MiB | <b>0.000</b> |
| Mean pooling      | 0.667     | 2.392           | 9.438        | 35.1 MiB        | 0.000        |
| Softmax attention | 0.720     | 3.125           | 12.729       | 652.1 MiB       | 0.842        |
| Self-attention    | 9.182     | <b>1128.474</b> | <b>OOM</b>   | 10381.6 MiB     | 1.960        |

**Table 1:** Latency and activation overhead.  $\alpha$  is the log-log exponent of overhead in  $N$ , fitted on  $N \geq 8192$ . Overhead excludes the input tensor, which is unavoidably  $\Theta(N)$  and would otherwise dominate.



**Figure 3:** Recall at 1% FPR,  $N=16,384$ , trained at  $N=512$ . Annealing converts a total training failure at  $S=0.10$  into full recall.

same architecture scores 0.00/0.00 at  $S=0.10$ : the gap between the two MultiMax rows in Fig. 3 is the entire contribution of §3.

### 5.3 Benchmark C: distributed attacks

An adversary who holds the total budget fixed but spreads it over  $m$  non-contiguous tokens presents per-token magnitude  $S/m$ , so

$$\max_j v_h^\top y_j = \Theta(S/m), \quad \text{while} \quad \frac{1}{N} \sum_j v_h^\top y_j = \Theta(S/N) \text{ is unchanged in } m, \quad (6)$$

the exact dual of Prop. 2.3. This is the attack aimed at a maximum, and it is the structural price of the hard reduction.

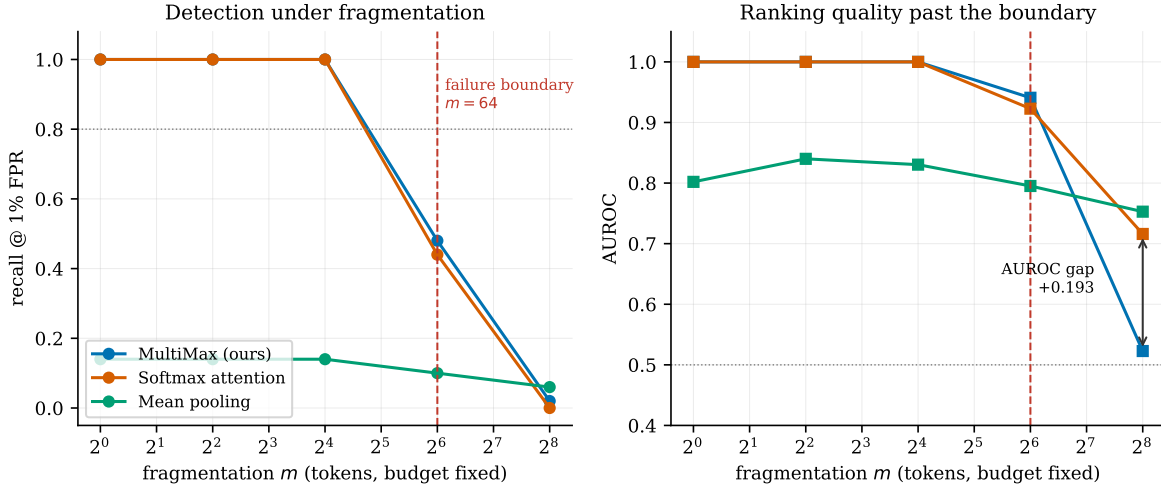
Two points, both against our own interest. First, the predicted MultiMax-specific vulnerability *did not isolate*: both pooled aggregators fail at the same  $m$ , so at this budget the distributed attack degrades query-based pooling as much as the hard reduction. Second, the recall boundary hides a real difference. At  $m=256$  both sit at recall  $\approx 0$  under a 1% FPR threshold, but AUROC is 0.523 for MultiMax against 0.716 for softmax: softmax keeps usable ordering after the thresholded detector has failed, while the hard maximum loses its signal more completely.

## 6 Related Work

**Probing and mechanistic interpretability.** Linear probes on intermediate representations date to Alain and Bengio [1]. Recent safety work trains them to detect deception [9] and evaluates them at scale [15, 17]. A consistent finding is that probes are strong in-distribution and brittle out of it [4, 20]. Parrack et al. [18] argue the right metric is the *black-to-white boost* over a black-box monitor, a standard we adopt.

| Strength $S$ | MultiMax (ours)    | Softmax attention | Mean pooling |
|--------------|--------------------|-------------------|--------------|
| 0.10         | <b>1.00</b> / 1.00 | 0.68 / 1.00       | 0.00 / 0.52  |
| 0.15         | 1.00 / 1.00        | 1.00 / 1.00       | 0.04 / 0.94  |
| 0.50         | 1.00 / 1.00        | 1.00 / 1.00       | 0.44 / 1.00  |

**Table 2:** Recall@16,384 / recall@512. The second entry separates “never learned the concept” from “learned it, then diluted”. Dilution ratio  $k/N = 2.4 \times 10^{-4}$ .



**Figure 4:** Fragmented attack: a fixed budget  $S=2.0$  split across  $m$  non-contiguous tokens. Both pooled aggregators fail at  $m=64$ ; past the boundary softmax retains markedly better ranking.

**Long-context guardrails.** Kramár et al. [14] deploy probes in a production assistant and report short-to-long generalisation failure with architecture as a governing factor. Chen et al. [7] attribute the failure to attention dilution and logit-margin compression. Our contribution is to isolate aggregation and give it a memory law.

**Sparse dictionaries.** Sparse autoencoders [16] are an alternative route to interpretable safety features, complicated by absorption [6] and by the finding that they need not beat random baselines [13]. We deliberately avoid dictionary dependence.

**Control and cascades.** AI-control protocols pair trusted and untrusted monitors [8, 10], and Koran et al. [12] show diverse monitor ensembles outperform homogeneous ones at equal compute. Our cascade is the low-latency first stage such protocols assume but rarely characterise. Calibration follows Guo et al. [11], Platt [19]; the smooth-max operator is that of Asadi and Littman [2]; the straight-through estimator is Bengio et al. [3].

## 7 Discussion, Rebuttal Analysis, and Limitations

**What we do and do not claim.** Combining Props. 2.3–2.4, Rem. 3.1 and §5.3: against mean aggregation, MultiMax wins unconditionally on concentrated attacks. Against single-query softmax pooling the advantage is conditional and narrow —  $\gamma$  in Prop. 2.4 is a *trained* quantity, and a salient needle drives  $e^\gamma \sim N/k$ , cancelling the dilution; post-annealing MultiMax leads only in the weak-signal regime and ties elsewhere. Against a distributed adversary the two fail together, with MultiMax degrading less gracefully. The unconditional advantage is therefore the systems one, and the deployment recommendation follows from it: use MultiMax as a  $\Theta(1)$ -memory first pass, not as a replacement for inspection.

| $m$ | per-token $S/m$ | MultiMax     | Softmax attention | Mean pooling |
|-----|-----------------|--------------|-------------------|--------------|
| 1   | 2.0000          | 1.000        | 1.000             | 0.140        |
| 16  | 0.1250          | 1.000        | 1.000             | 0.140        |
| 64  | 0.0312          | <b>0.480</b> | <b>0.440</b>      | 0.100        |
| 256 | 0.0078          | 0.020        | 0.000             | 0.060        |

**Table 3:** Recall at 1% FPR under fragmentation,  $N=16,384$ . Failure boundary (recall  $< 0.80$ ): MultiMax  $m=64$ , softmax  $m=64$ , mean pooling  $m=1$ .

**Rebuttal analysis.** Two objections shaped this paper. (i) “*Softmax outperforms MultiMax at  $m=256$ .*” Correct on AUROC (0.716 vs 0.523), and we report it in the abstract rather than the appendix; it is why our claim is architectural rather than a detection claim. (ii) “*Is 19.1 MiB truly  $\mathcal{O}(1)$ ?*” Half right, and it prompted the chunk ablation of Rem. 5.1, which replaces an  $\mathcal{O}(1)$  assertion with the measured law  $\Theta(\min(C, N))$ .

**Limitations.** *Synthetic activations.* Benchmarks run on Gaussian backgrounds with a single fixed additive attack direction; real misuse features are neither isolated nor axis-aligned, and the transfer of these results to real residual streams is untested. *Known- $m$  training.* Benchmark C trains on the same fragmentation it evaluates, which is generous to the defender; an unknown- $m$  adversary is untested. *Threshold drift.* Rem. 2.6 predicts  $\sqrt{2 \log N}$  growth in the benign maximum; our protocol recalibrates per length by construction and does not characterise the drift itself. *Simulated escalation.* The cascade’s second stage is a deterministic stand-in exercising routing and cost arithmetic; it is not evidence about a real monitor, and the cost constants are illustrative. *Single scale.* One width, one head count, one GPU.

**Reproducibility.** Every number in this paper is read from a single JSON artifact produced by the benchmark suite, and a checker verifies each quoted figure against it. This is not incidental: three figures in an earlier draft had silently gone stale after a bug fix changed the underlying model, and the checker exists because prose is not otherwise tested.

## References

- [1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.
- [2] Kavosh Asadi and Michael L. Littman. An alternative softmax operator for reinforcement learning. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.
- [3] Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*, 2013.
- [4] Gerard Boxo, Ryan Socha, Daniel Yoo, and Shivam Raval. Caught in the act: A mechanistic approach to detecting deception. *arXiv preprint arXiv:2508.19505*, 2025.
- [5] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [6] David Chanin, James Wilken-Smith, Tomáš Dulka, Hardik Bhatnagar, Satvik Golechha, and Joseph Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *arXiv preprint arXiv:2409.14507*, 2024.
- [7] Ziyang Chen, Xing Wu, and Songlin Hu. LongGuard: Mechanistic analysis and training-free mitigation of long-context failure in safety guardrails. *arXiv preprint arXiv:2608.27580*, 2026.
- [8] Nelson Gardner-Challis, Jonathan Bostock, Georgiy Kozhevnikov, Morgan Sinclair, Joan Velja, Alessandro Abate, and Charlie Griffin. When can we trust untrusted monitoring? a safety case sketch across collusion strategies. *arXiv preprint arXiv:2602.20628*, 2026.

- [9] Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim, and Marius Hobbhahn. Detecting strategic deception using linear probes. *arXiv preprint arXiv:2502.03407*, 2025. Apollo Research; ICML 2025.
- [10] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. *arXiv preprint arXiv:2312.06942*, 2023. Redwood Research.
- [11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.
- [12] Eugene Koran, Yejun Yun, Samantha Tetef, Benjamin Arnav, and Pablo Bernabeu-Pérez. Ensemble monitoring for AI control: Diverse signals outweigh more compute. *arXiv preprint arXiv:2605.15377*, 2026.
- [13] Anton Korznikov, Andrey Galichin, Alexey Dontsov, Oleg Rogov, Ivan Oseledets, and Elena Tutubalina. Sanity checks for sparse autoencoders: Do SAEs beat random baselines? *arXiv preprint arXiv:2602.14111*, 2026.
- [14] János Kramár, Joshua Engels, Zheng Wang, Bilal Chughtai, Rohin Shah, Neel Nanda, and Arthur Conmy. Building production-ready probes for gemini. *arXiv preprint arXiv:2601.11516*, 2026. Google DeepMind.
- [15] Kieron Kretschmar, Walter Laurito, Sharan Maiya, and Samuel Marks. Liars’ bench: Evaluating lie detectors for language models. *arXiv preprint arXiv:2511.16035*, 2026.
- [16] Tom Lieberum, Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma Scope: Open sparse autoencoders everywhere all at once on Gemma 2. *arXiv preprint arXiv:2408.05147*, 2024.
- [17] Vikram Natarajan, Devina Jain, Shivam Arora, Satvik Golechha, and Joseph Bloom. One probe won’t catch them all: Towards targeted deception detection. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *PMLR*, 2026.
- [18] Avi Parrack, Carlo Leonardo Attubato, and Stefan Heimersheim. Benchmarking deception probes via black-to-white performance boosts. *arXiv preprint arXiv:2507.12691*, 2026. Apollo Research.
- [19] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.
- [20] Daniel Yoo and Adrians Skapars. Probe generalization as subspace selection for OOD deception detection. *arXiv preprint arXiv:2609.02893*, 2026.